

Gaussian elimination

How to solve the standard problems of Lectures 0–4

Most computational problems of the first lectures are solved by one algorithm: write a matrix, reduce it to RREF, and read the answer off the pivots. The problems differ only in *which* matrix you write and *what* you read off. The elimination steps are written out once, in Lecture 0; here they are marked by a single $\xrightarrow{\text{RREF}}$ arrow. Each method ends with the sheet problems it solves.

Choosing the matrix

Vectors go into the *columns*; equations go into the *rows*. A bar appears only when the question names a target. Vectors of any other space are first replaced by their coordinate vectors in one fixed basis; for polynomials and matrices, the standard one (Example 2.15).

Lect.	Question	Matrix	Bar	Read-off
0	Solve $Ax = b$; the null space	equations as rows	$ b$	a row $[0 \cdots 0 c]$, $c \neq 0$: none; else x_p plus one vector per free variable (Section 1)
0	The inverse A^{-1}	A	$ I$	left half I : the right half; else no inverse (Section 1)
2	$v \in \text{span}(v_1, \dots, v_m)$? Coordinates $[v]_{\mathcal{B}}$?	the vectors as columns	$ v$	consistent? the solution (Section 2)
2	Independent? A basis of the span?	the vectors as columns, in order	—	the pivot columns (Section 3)
2	$U + W$ and $U \cap W$	bases of U , then W , as columns	—	pivots; relations (Section 4)
3	$\ker T$ and $\text{im } T$, $Tx = Ax$	A	—	free variables; pivot columns of A (Section 5)
4	Matrix of T , basis C of the codomain	C as columns	$ Tv_1, \dots, Tv_n$	the right half (Section 6)
4	Change of basis: $[v]_{\mathcal{B}'}$, A'	P	$ [v]_{\mathcal{B}};$ $ AP$	the right half (Section 6)

1 Linear systems

Lecture 0

Problem. Solve $Ax = b$: decide whether it has solutions, and write them all. For $b = 0$ this asks for a basis of the null space $\ker A$.

Method

1. Reduce $[A | b]$ to RREF. A row $[0 \cdots 0 | c]$ with $c \neq 0$: no solution.
2. Set every free variable to 0 and read off a particular solution x_p .
3. Take $b = 0$. For each free variable, set it to 1 and the other free variables to 0, and read off one vector. These vectors are a basis of $\ker A$.
4. The solution set is x_p plus any linear combination of the basis of $\ker A$.

Example. Solve the system $x_1 - 2x_2 + 2x_3 + x_4 = 2$, $3x_1 - 6x_2 + 5x_3 + 4x_4 = 7$:

$$\left[\begin{array}{cccc|c} 1 & -2 & 2 & 1 & 2 \\ 3 & -6 & 5 & 4 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} \textcircled{1} & -2 & 0 & 3 & 4 \\ 0 & 0 & \textcircled{1} & -1 & -1 \end{array} \right].$$

The free variables are x_2 and x_4 , and the pivot variables are $x_1 = 2x_2 - 3x_4 + 4$ and $x_3 = x_4 - 1$. Then:

$$\begin{aligned} x_2 = 0, x_4 = 0 &\implies x_p = (4, 0, -1, 0), \\ b = 0, x_2 = 1, x_4 = 0 &\implies u_1 = (2, 1, 0, 0), \\ b = 0, x_2 = 0, x_4 = 1 &\implies u_2 = (-3, 0, 1, 1). \end{aligned}$$

The general solution:

$$x = (4, 0, -1, 0) + s(2, 1, 0, 0) + t(-3, 0, 1, 1),$$

where s and t are arbitrary numbers. ◀

Check. Substitute x_p ; multiply A by each null-space vector.

Variant: the inverse. Reduce $[A \mid I]$. If the left half becomes I , the right half is A^{-1} ; a zero row on the left means A is not invertible.

Practice. Sheet 0: 0.22, 0.25, 0.26, 0.30, 0.31; 0.29 (split into cases before dividing by an expression in a); 0.23 (null space); 0.24, 0.27 (inverse). Sheet 2: 2.13, 2.16 (the solution set of a homogeneous system).

2 Span membership and coordinates

Lecture 2

Problem. Is v in $\text{span}(v_1, \dots, v_m)$, and with which coefficients? Coordinates $[v]_{\mathcal{B}}$ ask the same for a basis $\mathcal{B} = (v_1, \dots, v_m)$.

Method

1. Write $v_1, \dots, v_m$ into the columns and v after the bar; reduce.
2. A row $[0 \cdots 0 \mid c]$ with $c \neq 0$: v is not in the span.
3. Otherwise the solution gives the coefficients. For a basis it is unique: it is $[v]_{\mathcal{B}}$.

Example. Is $v = (7, 5, 8)$ in $\text{span}(v_1, v_2)$, where $v_1 = (1, -1, 2)$ and $v_2 = (3, 1, 4)$? We need numbers a and b such that $v = av_1 + bv_2$:

$$\left[\begin{array}{cc|c} 1 & 3 & 7 \\ -1 & 1 & 5 \\ 2 & 4 & 8 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} \textcircled{1} & 0 & -2 \\ 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 \end{array} \right] \implies a = -2, b = 3.$$

The last row is zero, so the system is consistent:

$$v = -2v_1 + 3v_2 \quad \text{and} \quad [v]_{\mathcal{B}} = (-2, 3).$$

Suppose instead that $v = (7, 5, 9)$. Then the RREF has the row $[0 \ 0 \mid 1]$, and $v \notin \text{span}(v_1, v_2)$. ◀

Practice. Sheet 2: 2.1–2.4 (span); 2.22–2.25 (coordinates).

3 Independence; reducing a list to a basis

Lecture 2

Problem. Is $v_1, \dots, v_m$ independent? If not, which vectors are redundant, and what are a basis and the dimension of the span?

Method

1. Write $v_1, \dots, v_m$ into the columns, in the order of the list; reduce.
2. The list is independent iff every column has a pivot.
3. A column without a pivot is redundant: its entries are its coefficients in the pivot columns to its left.
4. The pivot columns *of the original list* are a basis of the span; the dimension is the number of pivots.

Why step 4 works. Deleting the redundant vectors does not change the span (Lemma 2.10). The same row operations take the remaining columns to distinct standard vectors e_i , so they are independent.

Example. Are $v_1 = (1, 0, 2, 1)$, $v_2 = (2, 1, 3, -1)$, $v_3 = (3, 2, 4, -3)$, $v_4 = (0, 1, 1, 2)$ independent? Find a basis and the dimension of their span.

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 3 & 4 & 1 \\ 1 & -1 & -3 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & -1 & 0 \\ 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies v_3 = -v_1 + 2v_2.$$

So $v_1, \dots, v_4$ are dependent. The vectors v_1, v_2 and v_4 are independent and form a basis of $\text{span}(v_1, v_2, v_3, v_4)$, of dimension 3. ◀

Variants. Extend an independent $u_1, \dots, u_k$ to a basis of $\mathbb{F}^n$: reduce the list $u_1, \dots, u_k, e_1, \dots, e_n$ and keep the pivot columns. A *square* list with a parameter: the determinant (Example 2.9).

Pitfalls

- Taking the columns of the RREF as the basis: in general they do not even lie in the span.

Practice. Sheet 2: 2.6–2.10, 2.15, 2.19.

4 Sum and intersection of subspaces

Lecture 2

Problem. Given bases $u_1, \dots, u_k$ of U and $w_1, \dots, w_l$ of W , find the dimensions and bases of $U + W$ and $U \cap W$.

Method

1. Write $u_1, \dots, u_k$, then $w_1, \dots, w_l$, into the columns; reduce.
2. The pivot columns are a basis of $U + W$ (Section 3). Then $\dim(U \cap W) = k + l - \dim(U + W)$ (Theorem 2.32), the number of columns without a pivot.
3. Each column without a pivot gives a relation. Put its u 's on one side and its w 's on the other: both sides are one vector, lying in U and in W . These vectors are a basis of $U \cap W$.

Why step 3 works. The u 's carry pivots, so every column without one is some w_j , and w_j occurs in no other relation: the vectors are independent, and there are $\dim(U \cap W)$ of them (Proposition 2.24).

Example. Let $u_1 = (1, 2, 0, 1)$, $u_2 = (0, 1, 1, -1)$, $w_1 = (1, 1, 0, 2)$, $w_2 = (1, 2, -1, 1)$ and

$$U = \text{span}(u_1, u_2) \quad \text{and} \quad W = \text{span}(w_1, w_2).$$

Find the dimensions and bases of $U + W$ and $U \cap W$. We write u_1, u_2, w_1, w_2 into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 1 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 2 \\ 0 & \textcircled{1} & 0 & -1 \\ 0 & 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies w_2 = 2u_1 - u_2 - w_1.$$

Hence $\dim(U + W) = 3$, $\dim(U \cap W) = 2 + 2 - 3 = 1$, and

$$U + W = \text{span}(u_1, u_2, w_1), \quad U \cap W = \text{span}(2u_1 - u_2) = \text{span}((2, 3, -1, 3)),$$

since the vector $2u_1 - u_2 = w_1 + w_2$ lies in both U and W . ◀

Example. Let $u_1 = (1, 0, 1, 2)$, $u_2 = (1, 1, 0, 1)$, $u_3 = (0, 1, 2, 1)$, $w_1 = (2, 2, 0, 2)$, $w_2 = (0, 1, 1, 0)$, $w_3 = (2, 1, 0, 2)$ and

$$U = \text{span}(u_1, u_2, u_3) \quad \text{and} \quad W = \text{span}(w_1, w_2, w_3),$$

both of dimension 3. Find the dimensions and bases of $U + W$ and $U \cap W$. We write the six vectors into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 2 & 1 & 1 & 2 & 0 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & 2 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 1 \end{bmatrix}.$$

The fourth and sixth columns give $w_1 = 2u_2$ (so w_1 already lies in U) and $w_3 = u_1 + u_2 - u_3 + w_2$. Hence $\dim(U + W) = 4$, that is, $U + W = \text{span}(u_1, u_2, u_3, w_2) = \mathbb{R}^4$, and $\dim(U \cap W) = 3 + 3 - 4 = 2$:

$$U \cap W = \text{span}(2u_2, u_1 + u_2 - u_3) = \text{span}((2, 2, 0, 2), (2, 0, -1, 2)),$$

since $2u_2 = w_1$ and $u_1 + u_2 - u_3 = w_3 - w_2$ lie in both U and W . ◀

Pitfalls

- With the vectors in rows, as in Example 2.35, the RREF gives $\dim(U + W)$ but no relations, and so no basis of $U \cap W$.

Practice. Sheet 2: 2.27–2.30.

5 Kernel and image of a matrix map

Lecture 3

Problem. For $T: \mathbb{F}^n \rightarrow \mathbb{F}^m$, $Tx = Ax$, find bases and dimensions of $\ker T$ and $\text{im } T$; decide whether T is injective or surjective.

Method

1. Reduce the matrix A . There is no right-hand side, since it stays zero throughout.
2. *Kernel*: one vector per free variable, as for the null space in [Section 1](#); $\dim \ker T$ is the number of free variables.
3. *Image*: $\operatorname{im} T$ is spanned by the columns of A (Example 3.21), so reduce that list as in [Section 3](#): the pivot columns of A — not of its RREF — are a basis, and $\operatorname{rank} T$ is the number of pivots.

Example. Let $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be $Tx = Ax$. Find bases and the dimensions of $\ker T$ and $\operatorname{im} T$. We reduce the matrix A :

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 2 & 3 & -5 & 4 \\ -1 & 1 & -5 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 2 & -1 \\ 0 & \textcircled{1} & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies x_1 = -2x_3 + x_4, \quad x_2 = 3x_3 - 2x_4.$$

The free variables are x_3 and x_4 , and the pivot variables are x_1 and x_2 . Then:

$$\begin{aligned} x_3 = 1, \quad x_4 = 0 &\implies x = (-2, 3, 1, 0), \\ x_3 = 0, \quad x_4 = 1 &\implies x = (1, -2, 0, 1). \end{aligned}$$

Hence

$$\ker T = \operatorname{span}((-2, 3, 1, 0), (1, -2, 0, 1)), \quad \operatorname{im} T = \operatorname{span}((1, 2, -1), (2, 3, 1)).$$

Both have dimension 2: T is neither injective nor surjective. ◀

Check. Multiply A by each kernel vector.

Pitfalls

- ▶ The pivot columns of the RREF, $(1, 0, 0)$ and $(0, 1, 0)$, span the plane $x_3 = 0$, which does not contain $(1, 2, -1)$: row operations change the column space.
- ▶ $4 = 2 + 2$ is no check: pivots and free variables always add up to n .

Variant: a map from its values on a basis (Example 3.7). Find the coordinates $a, b, \dots$ of $(x, y, \dots)$ in that basis by [Section 2](#), with $(x, y, \dots)$ after the bar; then $T(x, y, \dots) = aTv_1 + bTv_2 + \dots$.

Practice. Sheet 3: 3.8, 3.10–3.12, 3.17, 3.22, 3.23, 3.29(a); 3.4 (a map from its values).

6 Several right-hand sides at once

Lecture 4

Problem. Find the matrix of a map in given bases, convert coordinates, or find the matrix $A' = P^{-1}AP$ of an operator in a new basis.

Method

1. Row operations never look at the right-hand side, so systems that share one matrix M are solved together. If M is invertible — as it is when its columns are a basis — $[M \mid B]$ reduces to $[I \mid M^{-1}B]$. If the left half does not become I , M is not invertible.
2. Choose M and B from the table below.

Question	M	B	Result
Inverse (Section 1)	A	I	A^{-1}
Matrix of T , basis C of the codomain	C as columns	$Tv_1, \dots, Tv_n$ as columns	$\mathcal{M}(T)$
New coordinates	P	$[v]_{\mathcal{B}}$	$[v]_{\mathcal{B}'}$
New matrix (Theorem 4.26)	P	AP	A'

All columns are coordinate vectors in one fixed basis of the codomain. P holds the new basis vectors as columns, written in the old basis (Definition 4.24). If the old basis $\mathcal{B} = (v_1, \dots, v_n)$ is not the standard one, first find P itself: reduce $[v_1 \cdots v_n \mid v'_1 \cdots v'_n]$ (all vectors in standard coordinates) to $[I \mid P]$. The other direction needs no reduction: $[v]_{\mathcal{B}} = P[v]_{\mathcal{B}'}$.

Example. An operator has the matrix $A = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}$ in the standard basis. Find its matrix A' in the new basis $(1, 1), (2, 3)$ of $\mathbb{R}^2$. Then $P = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $AP = \begin{pmatrix} 3 & 5 \\ 3 & 7 \end{pmatrix}$, and

$$\left[\begin{array}{cc|cc} 1 & 2 & 3 & 5 \\ 1 & 3 & 3 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|cc} \textcircled{1} & 0 & 3 & 1 \\ 0 & \textcircled{1} & 0 & 2 \end{array} \right] \Rightarrow A' = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

The right half is the matrix of the operator in the new basis. ◀

Example (A non-standard old basis). Let $\mathcal{B} = ((1, 1), (1, 2))$ and $\mathcal{B}' = ((2, 3), (1, 1))$. A vector has $[v]_{\mathcal{B}} = (1, 2)$, and an operator has $A = \mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Find $[v]_{\mathcal{B}'}$ and $\mathcal{M}(T, \mathcal{B}')$. First we find P :

$$\left[\begin{array}{cc|cc} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|cc} \textcircled{1} & 0 & 1 & 1 \\ 0 & \textcircled{1} & 1 & 0 \end{array} \right] \Rightarrow P = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $AP = \begin{pmatrix} 3 & 1 \\ 7 & 3 \end{pmatrix}$, and one reduction gives both answers:

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 3 \\ 1 & 0 & 2 & 7 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|cc} \textcircled{1} & 0 & 2 & 7 \\ 0 & \textcircled{1} & -1 & -4 \end{array} \right].$$

So $[v]_{\mathcal{B}'} = (2, -1)$ and $\mathcal{M}(T, \mathcal{B}') = \begin{pmatrix} 7 & 3 \\ -4 & -2 \end{pmatrix}$. ◀

Check. $PA' = AP$.

Pitfalls

- Reducing $[P \mid A]$ gives $P^{-1}A$, not A' : multiply by P first.

Practice. Sheet 4: 4.9, 4.11, 4.12 (if the left half does not become I , find a kernel vector by Section 5); 4.16–4.21; 4.10, 4.13 (Section 3, after coordinates).